PROJECT PULLI: Grounding Edge-Scale Pan-Indic Transliteration with Classical Phonotactics and In-the-Wild Golden Diagnostics

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Abstract

While large language models and commercial cloud APIs achieve competitive accuracy on formal dictionary transliteration, their performance degrades severely on organic, conversational typing across the Global South. In-the-wild inputs suffer from lexical translation leakage, fortis geminate collapse, and unconstrained cross-script hallucinations. PROJECT PULLI is an ultra-compact ($\approx$ 11.3M parameter / projected 11.3 MB INT8) edge-scale transliteration engine spanning 8 Indic languages (Tamil, Malayalam, Telugu, Kannada, Gujarati, Bengali, Hindi, and Marathi) trained on a 3.2M-pair Pan-Indic corpus. Grounding South Dravidian phonology in classical *Tolkāppiyam* phonotactics and building upon the empirical information-theoretic limits of benchmark text supervision [Lakshmanan, 2026], PULLI addresses the gap between clean benchmark corpora and noisy conversational chat across both Dravidian and Indo-Aryan scripts through a hybrid framework: pairing a Tier-1 Core Lexicon Trie with an edge-scale sequence-to-sequence Transformer regularised by language-adaptive objectives (chat augmentation and *puḷḷi* structural loss for Dravidian scripts; supervised contrastive spelling-invariance loss and nuqta preservation for Indo-Aryan). On a 5,000-sample macro benchmark, PULLI-A2 reaches **57.8%** versus 53.9% for a replicated INDICXLIT baseline (+3.9 pp, $+195$ net items; exact McNemar $p = 3.5 \times 10^{-14}$), and the language-routed system (**58.6%**) is non-inferior to Microsoft Azure at a 1.5 pp margin (58.4%; paired difference +0.22 pp, 95% CI $[-1.01, +1.45]$ pp). On an audited 216-sample in-the-wild golden suite verified by native speakers, PULLI achieves **71.8% Exact Match** (155/216 items), outperforming commercial cloud API Sarvam AI (66.7% EM, 144/216 items; +5.1 pp, $p = 0.21$) and open-weight INDICXLIT (67.6%; $p = 0.09$), both directional, while delivering approx. **19 ms per word on device** ($\approx 20\times$ faster than the cloud round trip) with zero infrastructure cost. PULLI establishes a statistically significant lead in Malayalam (+35.0 pp, $+14$ net items; exact McNemar $p = 0.00052$) alongside directional leads in Gujarati (+25.0 pp) and Telugu (+10.0 pp), while the commercial cloud API retains an advantage on Hindi, Marathi, and Bengali (-5.0 to -15.0 pp). The gains thus concentrate in, but are not confined to, Dravidian, and the split motivates the proposed language-adaptive routing, demonstrating that typologically grounded inductive biases offer a scalable and equitable blueprint for resource-efficient NLP.

1 Introduction

Machine transliteration across Indic scripts is fundamental for search, messaging, voice-input keyboards, and digital accessibility across South Asia and the wider Global South [Madhani et al., 2023,

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Roark et al., 2020]. Over 600 million users across the Indian subcontinent interact daily with digital platforms by typing native Dravidian and Indo-Aryan languages using the 26-letter Latin alphabet (*e.g.*, Tanglish, Manglish, Hinglish, Banglish).

However, existing commercial cloud services (*e.g.*, Sarvam AI Transliteration API; Sarvam AI, 2024) and large multilingual generative models exhibit critical failure modes when encountering authentic user chat across distinct language families:

1. **Task Confusion & Translation Leakage**: Commercial LLM-based APIs frequently confuse phonetic transliteration with semantic translation on everyday English loanwords (*e.g.*, Latin `acting` $\rightarrow$ Tamil *naṭippu* ["performance"] instead of spoken *ākṭiṅ*; `hospital` $\rightarrow$ Hindi *cikitsālay* instead of *hăspiṭal*).
2. **Lossy Single-Vowel Chat Collapse & Nasal Omission**: On mobile touchscreens, $> 90\%$ of users omit double vowels and diacritics, causing naive models to collapse grammatical markers (*e.g.*, Tamil `polama` $\rightarrow$ broken *polama* instead of modal *pōlāmā*; Hindi `hain` $\rightarrow$ *hai* instead of copula *haiṁ*).
3. **Voicing and Nuqta Errors**: Romanised chat writes post-nasal stops as voiced, and models read the Roman *b* as a phoneme (*e.g.*, Malayalam `kambani` $\rightarrow$ *kāṁbani*, with a lengthened vowel and anusvāra for the *mp* conjunct, instead of *kampani*); Perso-Arabic consonants lose their nuqta (Hindi `zindagi` $\rightarrow$ *jindagī*).
4. **High Latency & Cloud Compute Barriers**: Commercial cloud APIs incur round-trip network latencies exceeding 300 ms at non-zero operational costs, rendering them unviable for interactive keystroke-level IME typing on edge devices.

To bridge this gap across both linguistic families, this paper introduces **Project Pulli** (*puḷḷi* [*virāma*; Mal. *candrakkala*, Tel. *pollu*]). The primary contributions are fourfold:

- **Cross-Family Phonotactic Grounding**: For South Dravidian single-stop orthographies, classical *Tolkāppiyam* phonotactics are formalised—positional allophonic voicing of voiceless stops (*vallinam*) and fortis gemination via *puḷḷi* structural loss [Tolkāppiyar, c. 300 BCE – 200 CE, Niklas, 1988, Ramprashanth et al., 2025]; for Indo-Aryan scripts, Perso-Arabic nuqta preservation, vowel nasalisation augmentation, and latent spelling invariance are enforced.
- **Hybrid Edge-Scale Pipeline**: A Tier-1 Core Lexicon Trie for certified high-frequency anchors is paired with an 11.3M-parameter Transformer encoder-decoder trained on a 3.2M-pair Pan-Indic corpus and routed via language-adaptive objectives.
- **Two-Suite Benchmark Methodology**: PULLI-EVAL-5K, a 5,000-sample macro benchmark spanning 8 Indic scripts, is established alongside an audited 216-sample in-the-wild golden diagnostic suite verified by native speakers.
- **Competitive Edge Demonstration**: On the audited 216-sample suite, PULLI achieves **71.8% Exact Match** (155/216 items), outperforming commercial cloud API Sarvam AI (66.7% EM, 144/216 items; +5.1 pp directional, $p = 0.21$) while executing in approx. **19 ms per word** on device with zero compute cost, establishing a verified lead in Malayalam (+35.0 pp, $+14$ net items; exact McNemar $p = 0.00052$) alongside directional gains in Gujarati (+25.0 pp) and Telugu (+10.0 pp).

An extended discussion of related work is given in Appendix D.

2 System Architecture & Hybrid Pipeline

2.1 Hybrid Edge Pipeline & Language-Adaptive Routing

PULLI employs a two-tier hybrid inference pipeline designed for low-latency smartphone keyboards: (1) **Tier-1 Core Lexicon Trie**: Intercepts high-frequency conversational words (*e.g.*, greetings, modal pronouns, interrogatives) via an exact memory-mapped trie, returning certified canonical outputs via fast prefix lookup. (2) **Tier-2 Neural Sequence-to-Sequence Transformer**: Open-domain vocabulary, inflected nouns, and conversational phrases fall back to PULLI's sequence-to-sequence Transformer encoder-decoder ($\approx$ 11.3M parameters; Vaswani et al., 2017), regularised via phonological multi-task loss ($\mathcal{L}_{\text{phono}}$) and target-grounded chat augmentation [Lakshmanan, 2026]. To

Table 1: Top-1 Transliteration Accuracy (%) across 8 Indic languages on PULLI-EVAL-5K ($N = 5,000$) with 95% Wilson CIs. Gain (Δ) is PULLI-A2 minus replicated INDICXLIT-A0 in pp; PULLI-A3 results shown for architectural comparison. Overall $b = 430, c = 235$ (exact McNemar $p = 3.5 \times 10^{-14}$). *Statistically significant under family-wise Bonferroni correction ($\alpha = 0.00625$); †nominal ($\alpha = 0.05$). Complete 2×2 contingency cells are in Appendix C.

Language	Samples	IndicXlit-A0 (95% CI)	PULLI-A1	PULLI-A2 (95% CI)	PULLI-A3	Gain (Δ pp)
Tamil (`ta-IN`)	625	53.8% [49.8, 57.6]	53.3%	57.6% [53.7, 61.4]	**59.7%**	**+3.8**†
Malayalam (`ml-IN`)	625	54.4% [50.5, 58.3]	53.8%	59.2% [55.3, 63.0]	**62.6%**	**+4.8***
Telugu (`te-IN`)	625	64.3% [60.5, 68.0]	63.5%	**66.7% [62.9, 70.3]**	**66.7%**	+2.4
Kannada (`kn-IN`)	625	58.9% [55.0, 62.7]	61.8%	64.3% [60.5, 68.0]	**65.3%**	**+5.4***
Hindi (`hi-IN`)	625	47.8% [43.9, 51.8]	46.2%	51.5% [47.6, 55.4]	**52.3%**	**+3.7**†
Marathi (`mr-IN`)	625	54.1% [50.2, 58.0]	50.6%	**56.5% [52.6, 60.3]**	54.9%	+2.4
Bengali (`bn-IN`)	625	48.6% [44.7, 52.6]	47.0%	**53.3% [49.4, 57.2]**	48.8%	**+4.7***
Gujarati (`gu-IN`)	625	49.1% [45.2, 53.0]	49.0%	**53.1% [49.2, 57.0]**	48.8%	**+4.0**†
OVERALL (5,000)	5,000	53.9% [52.5, 55.3]	53.1%	**57.8% [56.4, 59.2]**	57.4%	**+3.9***

maximise accuracy across distinct linguistic typologies, PULLI's dispatcher dynamically routes requests by language family: **Dravidian Family (`ta`, `ml`, `te`, `kn`)** routes to PULLI-A3, trained with target-grounded lossy chat augmentation and *pulḷḷi* structural loss ($\mathcal{L}_{\text{pulḷḷi}}$); **Indo-Aryan Family (`hi`, `mr`, `gu`, `bn`)** routes to PULLI-A2, trained with supervised contrastive spelling-invariance loss ($\mathcal{L}_{\text{inv}}$), Perso-Arabic nuqta preservation, and vowel nasalisation augmentation.

3 Evaluation Framework

To ensure both large-scale statistical generalisability and fine-grained linguistic accountability, evaluation uses two suites (with the core lexicon trie deactivated during all benchmark and holdout evaluations to isolate pure neural generalisation; 0% trie hits across all reported tests): **Macro benchmark (5,000 stratified samples)**: The PULLI-EVAL-5K benchmark comprises 5,000 samples (625 per language across 8 scripts) categorised into Core Vocabulary (1,600 samples), Phonological Stress (1,600 samples), Multi-Word Contextual Phrases (1,000 samples), and Named Entities (800 samples). **Audited diagnostic suite (216 samples)**: To evaluate failure modes unrepresented in clean benchmark corpora, this study constructs an audited 216-sample in-the-wild golden suite spanning 8 Indic scripts (per-language counts in Table 2) probing five core stress typologies: (1) South Dravidian stop allophony; (2) lossy single-vowel chat collapse & nasal omission; (3) spoken inflectional contraction; (4) loanword translation leakage; and (5) Perso-Arabic nuqta preservation. Annotations were gathered via a collaborative spreadsheet distributed to native-speaker community researchers and linguists, instructing contributors to provide corrections matching phonetic Roman text regardless of formal dictionary canon (details in Appendix B).

4 Experimental Results

4.1 Macro Benchmark Performance

As detailed in Table 1, PULLI-A2 surpasses the INDICXLIT-A0 baseline across all 8 Indic languages, achieving an overall macro accuracy of **57.8%** (+3.9 pp, $+195$ net items of 5,000; exact McNemar $p = 3.5 \times 10^{-14}$; cells in Appendix C). Furthermore, family-adaptive champion routing (Section 2) achieves an overall macro accuracy of **58.6%** (+4.7 pp, $+235$ net items of 5,000 over baseline), establishing formal non-inferiority to commercial hyperscaler Microsoft Azure at a 1.5 pp margin (58.4%; paired difference +0.22 pp, 95% CI $[-1.01, +1.45]$ pp; exact McNemar $p = 0.75$, $b = 501$, $c = 490$). This family-level routing policy operates with a single degree of freedom (Dravidian $\rightarrow$ A3, Indo-Aryan $\rightarrow$ A2) motivated by linguistic typology; it deliberately forgoes 5 correct items on Hindi (where unrouted A3 scored 327 vs A2's 322). The progression separates three effects: (1) **A0 vs. A1**: Naive source-side multi-task phonological loss (PULLI-A1, 53.1%) yields no advantage over baseline (INDICXLIT-A0, 53.9%). In the Roman$\rightarrow$native direction the phonotactic context this auxiliary head supervises is already present in the source string the encoder reads, so the loss adds no information the model lacks; Lakshmanan [2026] likewise find no gain from phonological supervision with Transformer models. (2) **A2 vs. A3 Typological Trade-Off**: Coupling phonotactic priors with input-side chat augmentation (PULLI-A2, 57.8%) produces broad

Table 2: Native-Speaker Audited Golden Diagnostic Evaluation across all 8 Indic languages against Commercial Production Cloud API (Sarvam AI public REST API, August 2026). Exact item counts (k/n), 95% Wilson CIs, and lead (Δ) in pp are shown. *Statistically significant under two-tailed exact McNemar test ($p = 0.00052$, $b = 15, c = 1$). †Local inference compute on device; commercial cloud latency reflects client broadband roundtrip across 8 languages (mean 377.6 ms, 250 queries/language).

Language Script	Samples (n)	Cloud API (95% CI)	PULLI (95% CI)	Lead (Δ pp)
Dravidian Scripts				
Malayalam (`ml-IN`)	40	50.0% [35.2, 64.8] (20)	**85.0% [70.9, 92.9] (34)**	**+35.0***
Kannada (`kn-IN`)	15	80.0% [54.8, 93.0] (12)	80.0% [54.8, 93.0] (12)	+0.0
Telugu (`te-IN`)	20	75.0% [53.1, 88.8] (15)	**85.0% [64.0, 94.8] (17)**	**+10.0**
Tamil (`ta-IN`)	50	70.0% [56.2, 80.9] (35)	70.0% [56.2, 80.9] (35)	+0.0
Indo-Aryan Scripts				
Gujarati (`gu-IN`)	16	50.0% [28.0, 72.0] (8)	**75.0% [50.5, 89.8] (12)**	**+25.0**
Bengali (`bn-IN`)	20	**85.0% [64.0, 94.8] (17)**	80.0% [58.4, 91.9] (16)	−5.0
Hindi (`hi-IN`)	40	**60.0% [44.6, 73.7] (24)**	45.0% [30.7, 60.2] (18)	−15.0
Marathi (`mr-IN`)	15	**86.7% [62.1, 96.3] (13)**	73.3% [48.0, 89.1] (11)	−13.3
Pooled Audited Suite (216)	216	66.7% [60.1, 72.6] (144)	**71.8% [65.4, 77.3] (155)**	**+5.1**
User Latency (Roundtrip)	—	377.6 ms	**19.1 ms**†	≈ 20×
Cloud Compute Cost	—	$0.0015/req	**$0.00**	**Zero Egress**

gains across all scripts. Imposing target-grounded *pulli* structural loss (PULLI-A3) yields substantial empirical gains across South Dravidian scripts (Malayalam: **62.6%**, +3.4 pp over A2, +8.2 pp over baseline; Tamil: **59.7%**, +2.1 pp; Kannada: **65.3%**, +1.0 pp), but produces expected regressions on Indo-Aryan scripts lacking single-stop allophonic constraints (Bengali: −4.5 pp; Gujarati: −4.3 pp; Marathi: −1.6 pp). This confirms that phonological inductive biases are language-family bounded, validating family-specific champion routing.

4.2 Comparative Evaluation against Commercial Cloud APIs

Table 2 compares performance against commercial production APIs (Microsoft Azure Translator and Sarvam AI, August 2026; $n = 216$). Azure achieves 81.0% (175/216) exact match via web-scale vocabulary coverage on formal words ($p = 0.0045$, $b = 13$, $c = 33$ vs PULLI's 71.8%), while PULLI achieves exact parity on Tamil (70.0%) and close proximity in Malayalam (85.0% vs 87.5%). Against domestic cloud API Sarvam AI, PULLI achieves a +5.1 pp overall lead (71.8% vs 66.7%, +11 net items; directional, exact McNemar $p = 0.21$, $b = 37$, $c = 26$). Among the Dravidian languages, PULLI achieves a significant lead in Malayalam (**85.0%** vs 50.0%, **+35.0 pp**, +14 net items; exact McNemar $p = 0.00052$, $b = 15, c = 1$) and a directional lead in Telugu (+10.0 pp), with parity in Tamil (70.0%) and Kannada (80.0%). Within Indo-Aryan the picture splits: PULLI leads Gujarati (+25.0 pp, directional) while Sarvam leads Hindi (−15.0 pp), Marathi (−13.3 pp), and Bengali (−5.0 pp). On a 50-item per-language draw from the official AKSHARANTAR test split [Madhani et al., 2023], PULLI-A3 achieves **74.0%** EM in Tamil vs 40.0% (CER 4.16 vs 17.33; $b = 18, c = 1$, exact $p = 7.6 \times 10^{-5}$) and **90.0%** vs 24.0% in Malayalam (CER 1.91 vs 49.52; $b = 33, c = 0$, exact $p = 2.3 \times 10^{-10}$), outperforming Sarvam AI (pooled 82.0% vs 32.0%, +50.0 pp, $b = 51, c = 1, p = 2.4 \times 10^{-14}$) and the open-weight INDICXLIT-A0 baseline (82.0% vs 70.0%, $b = 15, c = 3$, $p = 0.0075$). Qualitative error studies are documented in Appendix G (Table 3).

5 Conclusion

PROJECT PULLI demonstrates that grounding edge-scale Transformers in classical phonotactics establishes formal non-inferiority to commercial cloud hyperscaler Microsoft Azure at scale (58.6% vs 58.4% on 5,000 samples at a 1.5 pp margin; $p = 0.75$) and achieves a directional lead over Sarvam AI on the audited suite (71.8% vs 66.7%, $p = 0.21$), significant in Malayalam (+35.0 pp, $p = 0.00052$), within an 11.3M parameter budget at approx. 19 ms per word on-device latency with zero cloud egress cost.

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A Hardware Latency Benchmarks & Test Protocols

To isolate compute efficiency from network transmission, inference latency was benchmarked across edge hardware and cloud environments:

- **On-Device Edge Compute (PULLI)**: Evaluated locally on Apple Silicon (Apple M4 Max, 64 GB unified memory) using the native PyTorch Metal Performance Shaders (MPS) execution runtime in single-batch evaluation ($N = 1$). Mean inference latency is **19.1 ms** per word ($\sigma = 6.9$ ms, median 18.0 ms, $p_{95} = 31.3$ ms; 1,080 single-word calls, batch size 1, eager PyTorch without compilation) at an unquantised model footprint of **45.2 MB** (projected to **11.3 MB** under post-training INT8 quantisation), well within interactive keystroke budgets.
- **Commercial Production Cloud API**: Evaluated by querying the public HTTPS REST endpoint (`https://api.sarvam.ai/transliterate`) from broadband network connections in South Asia and North America. Mean round-trip latency ranged from **358.6** to **404.2 ms** across the eight languages (**377.6 ms** overall; 250 requests per language; Table 2) and **333.5 ms** (100 requests) on the bilingual Aksharantar holdout suite. Network transit accounts for $> 95\%$ of end-to-end user wait time, precluding real-time interactive character streaming without recurring operational egress costs ($0.0015 per API call).

B Corpus Provenance, Lexicon Grounding & Annotation Guidelines

B.1 Training Corpora & Multi-Corpus Pan-Indic Grounding

PULLI's 3.2M-pair Pan-Indic training corpus was constructed through multi-corpus grounding across 8 languages, explicitly distinguishing formal dictionary sources from colloquial conversational corpora:

- **Pan-Indic Core Corpora**: All 8 Indic languages (Tamil, Malayalam, Kannada, Telugu, Gujarati, Bengali, Hindi, and Marathi) draw upon verified lexical pairs from the DAKSHINA dataset [Roark et al., 2020], AKSHARANTAR [Madhani et al., 2023], AI4Bharat INDICCORP regional press text [Kakwani et al., 2020], and Wikimedia dumps.
- **Conversational Social Media Grounding**: For South Dravidian conversational chat, uncurated social comments were derived from the FIRE 2020 DravidianCodeMix corpus [Lakshmanan and Ravindranath, 2020] covering Tamil and Malayalam YouTube conversational threads. For the remaining six languages, conversational chat patterns were extracted from Wikimedia talk pages, subtitle dialogues, and colloquial messaging corpora.
- **Dravidian & Indo-Aryan Etymological Paradigms**: To prevent out-of-vocabulary collapse on rich agglutinative and inflected forms, systematic verbal and cognate paradigms were grounded in the *Dravidian Etymological Dictionary* [DEDR; Burrow and Emeneau, 1984] for Dravidian languages (Tamil, Malayalam, Kannada, Telugu) and Turner's *Comparative Dictionary of the Indo-Aryan Languages* [CDIAL; Turner, 1962–1966] for Indo-Aryan languages (Hindi, Bengali, Gujarati, Marathi).

- **Tier-1 Pragmatic Conversational Anchors**: The Tier-1 Core Lexicon Trie intercepts closed-class functional categories—greetings, discourse markers, deictic particles, interrogatives, and modal auxiliaries. Even under zero-cost caching, character-level generative sequence models are sensitive to small input perturbations on vowel-omitted conversational tokens, whereas trie lookup returns the certified form exactly with 100% canonical reproducibility for core interactive keystrokes.

B.2 Evaluation Benchmarks & Diagnostic Annotation Protocol

While PULLI-EVAL-5K was programmatically stratified from verified conversational dialogue and subtitle corpora (gated against master Indic lexicons), the 216-sample golden diagnostic suite was curated to capture adversarial conversational edge cases:

- **Golden Suite Annotator Recruitment & Adjudication**: Annotations were gathered from 8 native-speaker community researchers and linguists across Tamil, Malayalam, Kannada, Telugu, Gujarati, Bengali, Hindi, and Marathi communities via unpaid voluntary participation. Annotators were provided with uncurated Romanised chat queries in a shared collaborative sheet with the verbatim instruction: *"Please add a correction to the Indic text if the default transliteration is wrong. It need not be a dictionary word, but it should match the phonetic Roman text. If someone else has added a correction and you want to add one more, please add it in a new column."* When annotators documented valid phonetic or dialectal variants (*e.g.*, Malayalam `kollam` → *kollaṁ* / *koḷḷāṁ*; Hindi `kyun` → *kyōṁ* / *kyūṁ*), both variants were retained in the gold reference set (adjudication by union of accepted variants). A prediction is scored as correct if it matches any approved native target. Malayalam references mix two spellings of the word-final half-*u*, the modern *candrakkala* and the pre-reform *u* vowel sign, and Roman `-chu` is ambiguous between past tense and conjunctive participle; exact match scores either mismatch as an error, which costs each system at most 2.6 pp in Malayalam on the 5,000-item benchmark (PULLI-A3: 1.8 pp). The reference set is left unchanged.
- **Train/Test Contamination & Trie Deactivation Audit**: A full scan of all 52.1M training lines (26.6M unique pairs) reveals minimal pair-level overlap on PULLI-EVAL-5K (2.1%, 103/4,993). On the 216-item audited suite, partitioning performance by seen ($n = 110$) versus unseen ($n = 106$) training pairs demonstrates that PULLI-A3 does not degrade on unseen items (70.0% seen vs 73.6% unseen; $+3.6$ pp), while Microsoft Azure drops by -7.2 pp (84.5% to 77.4%) and Sarvam drops by -3.1 pp (68.2% to 65.1%, identical to INDICXLIT's -3.1 pp drop from 69.1% to 66.0%). PULLI's lead over Sarvam expands from $+1.8$ pp on seen pairs ($b = 16, c = 14$) to $+8.5$ pp on genuinely unseen pairs ($b = 21, c = 12, p = 0.16$). Crucially, on the 100-item public AKSHARANTAR holdout split, pair-level contamination is strictly 0.0% (0/50 Tamil, 0/50 Malayalam), where PULLI-A3 achieves 82.0% vs 32.0% for Sarvam ($+50.0$ pp, $b = 51, c = 1, p = 2.4 \times 10^{-14}$) and 70.0% for INDICXLIT-A0 ($b = 15, c = 3, p = 0.0075$). Furthermore, to prevent memorisation artifacts and isolate genuine neural generalisation, the Tier-1 Core Lexicon Trie was 100% deactivated during all macro (5,000-sample), golden diagnostic (216-sample), and AKSHARANTAR holdout (100-sample) evaluations (0% trie hits across all reported benchmarks); all reported benchmark scores reflect unassisted sequence-to-sequence Transformer inference.
- **Licensing & Data Availability**: Both benchmarks, alongside evaluation scripts and caching harnesses, will be released under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence upon publication.

C Statistical Hypothesis Testing & 2×2 Contingency Matrices

Table 1 evaluates the macro performance of PULLI-A2 against replicated INDICXLIT-A0 across all 5,000 benchmark samples. For paired McNemar testing, the complete 2×2 contingency cells (a, b, c, d)—where a is both models correct, b is correct only under A2, c is correct only under A0, and d is neither correct—are documented below:

- **Kannada (`kn-IN`)**: $a = 344, b = 58, c = 24^*, d = 199$ (exact $p = 2.2 \times 10^{-4}$)
- **Malayalam (`ml-IN`)**: $a = 310, b = 60, c = 30^*, d = 225$ (exact $p = 0.0021$)

- **Bengali (`bn-IN`)**: $a = 275, b = 58, c = 29^{*}, d = 263$ (exact $p = 0.0025$)
- **Gujarati (`gu-IN`)**: $a = 277, b = 55, c = 30^{\dagger}, d = 263$ (exact $p = 0.0088$)
- **Tamil (`ta-IN`)**: $a = 306, b = 54, c = 30^{\dagger}, d = 235$ (exact $p = 0.0116$)
- **Hindi (`hi-IN`)**: $a = 262, b = 60, c = 37^{\dagger}, d = 266$ (exact $p = 0.025$)
- **Telugu (`te-IN`)**: $a = 374, b = 43, c = 28, d = 180$ (exact $p = 0.096$)
- **Marathi (`mr-IN`)**: $a = 311, b = 42, c = 27, d = 245$ (exact $p = 0.091$)
- **Pooled Benchmark ($N = 5,000$)**: $a = 2459, b = 430, c = 235^{*}, d = 1876$ (exact $p = 3.5 \times 10^{-14}$)

*Statistically significant under family-wise Bonferroni correction ($\alpha = 0.00625$). †Significant at nominal level ($\alpha = 0.05$).

D Extended Related Work

Indic Language Technologies: The AKSHARANTAR benchmark and multilingual INDICXLIT baseline [Madhani et al., 2023] established large-scale transliteration baselines across 21 Indic languages.

In-the-Wild Romanisation Corpora: Research on code-mixed Dravidian social media text established that noisy Romanised user text exhibits severe phonetic distortion requiring specialised feature handling [Lakshmanan and Ravindranath, 2020]. Inductive biases and tokenisation strategies across morphologically rich Indic languages were investigated by Vemula et al. [2025]. The limits of phonological supervision on benchmark text and its contrast with spoken speech acoustics were studied by Lakshmanan [2026].

E Typological Scope of Classical Phonotactic Priors across Dravidian Languages

We delineate the precise linguistic and typological boundaries of classical phonotactic priors across the four evaluated Dravidian languages:

1. **Tamil and Malayalam (Stop-Voicing Allophony)**: The *Tolkāppiyam* (*Eḻuttatikāram*; c. 300 BCE–200 CE; Tolkāppiyar, c. 300 BCE – 200 CE) codifies Old Tamil phonology, establishing a single series of voiceless plosives (*vallinam*: $k, c, \underline{t}, t, p, \underline{r}$) operating under strict positional allophony (voiceless word-initially and when geminated, but voiced post-nasally and lenited intervocalically; Niklas, 1988, Ramprashanth et al., 2025). Because Malayalam historically diverged from Western Middle Tamil between the 9th and 13th centuries CE [Ramaswami Ayyar, 1936, Krishnamurti, 2003], its native vocabulary, together with early nativised loans (*tadbhava*), inherits this complementary voicing distribution. While Malayalam came to be written in the Grantha-derived *Ārya-eḻuttŭ* script, whose voiced and aspirated letters serve Sanskrit borrowings (*tatsama*), words within the native phonological system, including nativised loans, keep the single-stop allophonic distribution: English *company* is written *kampani* and pronounced [kambani], the voiceless grapheme *pa* voiced after the nasal.
2. **Kannada and Telugu (Contrastive Voicing vs. Virāma Regularisation)**: Kannada (South Dravidian) and Telugu (South-Central Dravidian) possess independent epigraphic and literary grammatical traditions (*Kavirājamārga*, Keśirāja's *Śabdamaṇidarpaṇa*, and *Āndhra Śabda Cintāmaṇi*; Krishnamurti, 2003, Zvelebil, 1990). Both languages feature phonemically contrastive voiced stops with dedicated graphemes (e.g. Kannada *ka* vs *ga*; Telugu *ka* vs *ga*). We do **not** claim that *Tolkāppiyam* stop-voicing rules govern Kannada or Telugu inventories. Rather, what unifies the Dravidian champion routing (PULLI-A3) across all four scripts is **virāma (*pulḷi* [*candrakkala / pollu*]) structural regularisation ($\mathcal{L}_{\text{pulḷi}}$)**, which penalises broken ligatures and supervises fortis geminates (*dvitva*) against spurious vowel insertions in conversational mobile typing, alongside target-grounded vowel length reconstruction.

F Why Source-Side Phonological Supervision Does Not Help

PULLI transliterates from Roman to native script, so stop voicing is carried by the input: voiced and voiceless spellings of a native single stop map to the same output grapheme (e.g. `thambi` and `thampi` both to *tampi*). An auxiliary head that predicts phonotactic context from the source string therefore supervises information the encoder already reads, which is consistent with PULLI-A1 (53.1%) not improving on the unregularised baseline (INDICXLIT-A0, 53.9%); Lakshmanan [2026] likewise find no gain from phonological supervision with Transformer models. What does differ between benchmark and conversational text is the input distribution: benchmark romanisations follow elicited spelling conventions, whereas conversational typing is more often phonetic (*e.g.* `thangam`, `kambani`). Input-side chat augmentation (PULLI-A2, 57.8%; $p = 3.5 \times 10^{-14}$) and target-grounded *pu̱l̤l̤i* structural loss (PULLI-A3, 57.4%; $p = 1.2 \times 10^{-11}$) address that mismatch, and that is where the significant gains appear.

Use of AI-Assisted Technologies

During the preparation and execution of this research, the author utilised AI-assisted systems for specialised research workflows: Anthropic Claude was utilised during initial system design ideation and as an automated peer reviewer providing critical structural critique on working drafts; Google Antigravity / Gemini was utilised as an agentic pair-programming assistant for implementing data processing scripts, orchestrating PyTorch benchmark runs, compiling diagnostic scorecards, and formatting LaTeX typography and bibliography assets. The author conceived the research architecture, directed the experimental methodology, verified all empirical benchmark data, authored the manuscript, and takes full scientific and legal responsibility for the integrity and conclusions of this publication.

G Qualitative Mechanical Error Case Studies

Table 3 provides qualitative side-by-side error case studies demonstrating typical mechanical failure classes encountered in conversational user text.

Table 3: Qualitative Mechanical Error Case Studies on Conversational Stress Probes. Wild user-typed inputs are quoted literally in `typewriter`; model predictions are given in standard ISO 15919 scientific transliteration with literal phonetic glosses.

Language	Wild Roman Input	Challenge Phenomenon	Commercial Production API	PROJECT PULLI
Malayalam	`kambani`	Post-Nasal Voicing & Conjunct	*kāṁbani* *(Lengthened ā, voiced ba, anusvāra for the mp conjunct)*	***kampani*** [kambani] *(written mp, pronounced [mb])*
Malayalam	`manichitrathazhu`	Retroflex Liquid & Cluster	*maṇiccayuttal* *(Severe token collapse on tra and ḻu)*	***maṇicittrattāḻ*** *(Preserves conjunct ttra and retroflex coda ḻ)*
Kannada	`kantara cinema`	Double Long Vowel Retention	*kātara sinimā* *(Nasal anusvāra erased: kāntāra → kātara)*	***kāntāra sinimā*** *(Preserves long ā and homorganic nasal nta)*
Tamil	`theedhum nandrum`	Positional Voicing of *vallinam* Stops	*tētum nantarum piṟaṟa* *(Corrupted vowels and stops; hallucinates alveolar ṟṟ)*	***tītum naṉṟum*** (*theedhum nandrum*) *(Faithful post-nasal vallinam voicing and initial voicelessness in classical register)*
Tamil	`verithanamana`	Fortis Gemination (*pu̱l̤l̤i*)	*veṟitaṇamāṇa* *(De-geminated: missing pu̱l̤l̤i over plosive t)*	***veṟittaṇamāṇa*** *(Enforces fortis gemination via pu̱l̤l̤i loss $\mathcal{L}_{pu\underline{l}\underline{l}i}$)*
Gujarati	`kemcho tame maja`	Conversational Greeting	*kēñcō tamē majāmāṃ* *(Corrupted nasalisation)*	***kēm chō tamē majā mā*** *(Clean colloquial greeting tokens)*
Hindi	`zindagi safar`	Perso-Arabic Nuqta	*zindagī saphar* *(Erroneous double nuqta / aspiration)*	***zindagī safar*** *(Standard Perso-Arabic fricative nuqta)*